

From Semantic Induction Heads to Multi-hop Circuits

Research Progress and Experiment Record

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Working record — updated after the OLMo-2-7B comparison (September 2026)

1 Scope and Current Status

Research log for the project, from proposal to the model-platform decision. It is a record for writing the final report, not the report itself.

Completed: literature study and methodology design; synthetic multi-hop datasets with built-in controls (four major revisions); evaluation infrastructure (multi-GPU Slurm, resumable runs, manifests, integrity checks); behavioral pilots on Pythia-1B and Pythia-6.9B; an order-pair analysis that separates content-based from layout-based success; a tokenizer-matched comparison run on OLMo-2-7B with a pre-registered decision rule; and the platform decision: **OLMo-2-7B is the primary model for the mechanistic phase, Pythia-6.9B is retained as a contrast model.**

Not yet done: Relation Index (RI) measurements on our models, activation or path patching, circuit extraction, faithfulness evaluation, checkpoint sweeps. No Semantic Induction Circuit (SIC) has been identified yet.

2 Research Question

From the proposal (*From Heads to Circuits: Identifying Semantic Induction Circuits for Multi-hop Reasoning in LLMs*):

How do language models process multi-hop semantic relations in context? Are these relations handled by individual semantic induction heads or by distributed circuits across layers?

Concretely: identify the circuit responsible for two-hop in-context composition (e.g. $A \rightarrow B$, $B \rightarrow C \Rightarrow A \rightarrow C$ over invented entities), validate it causally, and track its emergence across public training checkpoints. The behavioral phase below established *which model* reliably performs the task and *what* the circuit must explain (including its characteristic failures); the edge-rewiring conditions double as the corrupted inputs that patching requires.

3 Literature and Method Stack

Core sources: the transformer-circuits framework (residual stream, QK/OV, head composition) [3]; induction heads and their role in ICL [2]; semantic induction heads and the Relation Index, the direct starting point [1]. Causal methodology: the IOI circuit and path patching [4]; activation-patching practice [5]. For circuit discovery at scale we selected position-aware edge attribution (PEAP-style schemas suit our templated prompts) with gradient-based edge scoring (EAP-IG) for screening, manual activation/path patching for verification, and a pre-registered faithfulness

criterion. Two standing cautions from the reading: an SIH label does not establish the classical copying circuit (separate prefix-matching/copying controls are planned), and RI is observational — a screening signal, not causal evidence.

4 Planned Mechanistic Pipeline

1. **Behavioral validation** (done): direct retrieval, two-hop composition, edge rewiring, robustness; order-pair design.
2. **Candidate screening**: RI (with random-target null), attention patterns, per-component logit attribution. Compute order matters: apply the QK dominance filter from cached attention first, and project through OV to the vocabulary only for surviving (head, position) pairs at entity positions — the naive all-heads $\times$ all-positions OV projection costs ~ 30 forward passes per prompt.
3. **Node interventions**: activation patching between clean and rewired prompts; effect measured on the answer margin

$$m(x) = \log P(y_{\text{target}} \mid x) - \log P(y_{\text{alternative}} \mid x),$$

with the three-way logit contrast (correct new answer / new bridge / old answer) as the finer-grained outcome.

4. **Directed connections**: path patching to substantiate “ h_1 feeds h_2 ”; Q/K vs V effects separated.
5. **Circuit validation**: faithfulness of the sparse subgraph (recovery threshold fixed in advance), held-out families, generalization across the two fact orders.
6. **Checkpoint sweep**: repeat screening plus a fixed patching battery on ~ 10 – 15 OLMo-2-7B revisions; a *costing dry-run* on the final checkpoint (time per cached forward, per full RI pass, per patched forward; backward-pass memory fit for gradient attribution on 11–12 GB GPUs) precedes any sweep commitment.

5 Models, Data, and Infrastructure

Models (base LMs, no fine-tuning; ICL only; fp16; revisions pinned in per-model lock files):

- Pythia-1B: f73d7dcc545c8bd326d8559c8ef84ffe92fea6b2;
- Pythia-6.9B: c0e3eee36dc47af0c49f361c74cfe459c09f7f23;
- OLMo-2-1124-7B: 7df9a82518afdecae4e8c026b27adccc8c1f0032 (weights stored fp32, 27.2 GiB on disk; run in fp16, ≈ 14.6 GiB across two 11–12 GB GPUs — the same infrastructure as Pythia-6.9B).

Compute: TAU Slurm (`studentkillable`); node s-002 excluded (faulty GPU 0).

Data design. Synthetic families with invented names: a query chain and a distractor chain per family (kinship anchor world: child $\rightarrow$ mother $\rightarrow$ grandmother; breadth worlds: containment, location). Gold relations known by construction; an independent verifier re-solves every question from prompt text alone. Controls built into the data: mention balance (frequency rule scores exactly 50%), candidate token-length matching, no options line (candidates scored outside the prompt), demo/test name disjointness, and two fact-order variants per question.

Runner. `pilot_v2/pilot.py + parallel.py`: multi-GPU sharded evaluation by family, resumable, manifests with model/data/code hashes. Now multi-model (`-model {pythia-6.9b, olmo2-7b}`); each dataset row records the tokenizer(s) it was length-matched for, and the runner *refuses* a model/dataset combination that was not audited together. Numerical setup: fp16 weights, math-only SDPA, explicit finite-value checks (adopted after non-finite logits in early 6.9B runs).

6 Evaluation Measures

Candidate accuracy: the correct name outscores one matched distractor by summed log-probability of the full continuation (leading space and period included; tokenization boundary asserted). **Free accuracy**: the first generated name equals gold. **Order-pair (both-orders) accuracy** — the primary robustness measure: each question appears in two versions differing only in whether the query family’s facts are far from or adjacent to the question; a question counts only if answered correctly in *both*. Chance is 25%, and a model that answers by position fails at least one version by construction. Families are the statistical unit for interval estimates. Free-generation errors are categorized (bridge entity / distractor / other) rather than only counted.

7 Experiment History (Pythia)

7.1 Yes/No formats: a dead end (Pythia-1B, then 6.9B)

The initial entailment task (Yes/No over graph-derived facts) failed at both scales: the 1B model was label-biased (88% on positives, 19% on negatives); the 6.9B model answered “No” almost universally (1,154–1,200 of 1,200) across two demonstration revisions, including a balanced one (`prompt-v3`). Conclusion, consistent with [1] (App. D): base LMs of this scale do not comply with verification formats. All later work uses name completion.

7.2 Completion-v1: real one-hop signal, confounded two-hop

Candidates shown in the prompt; 1,200 questions per shot condition. Direct retrieval reached 91% (candidate, 4-shot), but the two-hop result was uninterpretable: the correct candidate was usually the more frequently mentioned one (a frequency rule scored 100% on basic two-hop), candidate position in the options line biased answers, and half of the free two-hop answers named the bridge entity. Changed-fact pair success was 3–19%. Verdict: test flaws, not measured ability.

7.3 Completion-v2: over-corrected into a negative result

Rephrasing as explicit graph navigation (“shortest chain of exactly k steps”) with fully balanced candidates collapsed *everything* to $\approx 50\%$, including one-hop. Retained as a documented negative result: base LMs are highly sensitive to unnatural, meta-linguistic prompt formats. Fix confounds in data structure and scoring; keep surface language natural.

7.4 Completion-v3 / v3.1: the current instrument

Natural language, no preamble, no options line; the composed relation has its own name (*grand-mother*), so the bridge is never a valid answer in the anchor world; mention balance and token-length matching by construction; 100 families (68 kinship, 16 containment, 16 location) $\times$ 6 questions $\times$ 2 fact orders = 1,200 prompts per shot condition (0/4/12).

v3’s first run contained a demonstration confound we caught by the order split: demos coupled hop count to fact-block order and taught a position rule (two-hop 34–50% in one order vs 93–97% in the other). **v3.1** fully crosses hops $\times$ order in the demonstration bank. All headline results below are v3.1.

Pythia-6.9B on v3.1 — two regimes (kinship, candidate accuracy, order_{far}/order_{adjacent}/both, 12-shot): direct 0.84/1.00/0.84; two-hop 0.72/1.00/0.72 (above chance in both orders, but fully asymmetric); *reorder* (shuffled facts) 0.51/0.41/0.25 — exactly chance; *broken/first* (first edge rewired to the other family’s mother) 0.18/0.00/0.00 — systematically wrong, dominated by the *old* grandmother; *broken/second* 0.68/1.00/0.68 — tracked correctly. Interpretation (hypothesis): hop-2 (bridge→answer) is entity-based; hop-1 (query→bridge) is resolved by layout/locality, which the canonical layout happens to align with truth.

8 Model Comparison: OLMo-2-7B

8.1 Selection

To test whether the layout-bound regime is a training-data limitation (Pythia: ~ 0.3 T tokens) rather than a scale limitation, we compared at matched size against OLMo-2-1124-7B (~ 4 T tokens; same 32 layers $\times$ 32 heads). Larger OLMo variants were rejected: 13B (≈ 27 GB fp16) needs 3–4 GPUs per job and 32B (≈ 64 GB) is infeasible on the partition; TransformerLens supports the 1B and 7B OLMo-2 models but not 13B/32B; and the analysis phase consists of many small jobs, where per-job GPU count dominates queue time. OLMo-2 also publishes intermediate checkpoints (`stepXXX-tokensYYYB` revisions), which the later sweep requires.

8.2 Tokenizer audit and adapted dataset

v3.1 candidate pairs were length-matched under Pythia’s tokenizer; under OLMo-2’s (~ 100 K vocabulary) 1,450 of 3,640 rows (62 families) broke — a direct scoring bias if uncorrected. `pilot_v3/adapt_data_olmo2.py` renamed 81 entities in 60 families (anchored on each family’s two-hop gold; replacements length-matched under *both* tokenizers and absent from all prompts). Everything else — ids, question structure, demonstration blocks (byte-identical), mention balance — is unchanged. Verbatim controls use demonstration names and are flagged as unmatched by design. Verified: pair-length and boundary audits clean under both tokenizers; logic verifier 0 errors on 3,600 questions and 19,200 demonstration blocks. A preflight script re-ran the audit on the cluster with the downloaded snapshot tokenizer before any GPU job.

8.3 Results

Pipeline calibration (OLMo controls): verbatim 20/20 candidate; stated 15/20, all five failures selecting the more *recent* distractor fact over the explicitly stated answer (model behavior — recency competition — not a pipeline defect). A Pythia run of the same controls file is still pending.

Kinship, candidate accuracy, order_{far}/order_{adjacent}/**both**, 12-shot:

Check	Pythia-6.9B	OLMo-2-7B
direct	0.84/1.00/ 0.84	1.00/0.90/ 0.90
two-hop	0.72/1.00/ 0.72	0.82/0.88/ 0.78
reorder	0.51/0.41/ 0.25	0.62/0.57/ 0.37
broken/first	0.18/0.00/ 0.00	0.15/0.44/ 0.09
broken/second	0.68/1.00/ 0.68	0.94/0.91/ 0.85

Three findings:

1. **Order symmetry restored.** OLMo’s two-hop is 0.82/0.88 vs Pythia’s 0.72/1.00: success no longer depends on where the query family sits.
2. **Above chance on shuffled facts — first such evidence.** Reorder both-orders: OLMo 0.41 [Wilson 0.30, 0.53] at 4-shot ($p = 2.5 \times 10^{-3}$ vs chance 0.25) and 0.37 [0.26, 0.49] at 12-shot; Pythia is exactly at chance ($p = .75/.55$).
3. **Same broken/first failure rate, different failure.** Free-generation anatomy (kinship, 68 questions): Pythia’s dominant error is the *old* grandmother (rewired edge ignored — the layout answer); OLMo’s is the *new* bridge (rewired edge followed by content, composition incomplete). Shot dependence for OLMo: correct-new-answer 33 (0-shot) $\rightarrow$ 13 (4) $\rightarrow$ 6 (12); bridge 25 $\rightarrow$ 38 $\rightarrow$ 41. **Well-formed demonstrations actively suppress OLMo’s completion of the rewired chain** — a demo-induced prior competing with in-context evidence.

Caveats: broken cells are $n=34$ order pairs; per-world two-hop (both-orders, 12-shot) is kinship P.72/O.78, location P.12/O.69, but containment P.44/O.25 — OLMo is *worse* there (format interaction; containment must not be pooled with kinship); OLMo is somewhat more distractible in the adjacent order (distractors variant 0.72 vs Pythia 0.91).

8.4 Pre-registered decision rule and outcome

Fixed before any OLMo number: adopt OLMo iff (1) direct and two-hop at least Pythia’s level; (2) broken/first above 50% in both orders; (3) broken/second at least Pythia’s level; secondary: reorder above chance. Outcome at 12-shot: (1) pass, (2) **fail**, (3) pass, secondary pass — a partial pass. **Documented deviation:** the rule’s failure clause assumed a Pythia-like (layout) failure; the observed failure is a composition bottleneck on top of working content-based binding — the mechanism the project needs. Decision: OLMo-2-7B primary, Pythia-6.9B retained as a mechanistic contrast (two regimes on identical data strengthen, not weaken, the comparison).

8.5 Candidate mechanisms for the demonstration-suppression effect

Status: fact = the output distribution over {new answer, new bridge, old answer} shifts causally with shot count; everything below is hypothesis. (i) *Truncated composition*: hop-1 succeeds on the rewired edge; the completing fact (far in one order) is not retrieved — supported by bridge answers concentrating where that fact is far (25 vs 16 of 34). (ii) *Routine confusion*: demos contain both one-hop and two-hop questions; the bridge is the correct answer to the wrong (one-hop) question. Cheap discriminating experiment: demo banks of only-mother vs only-grandmother questions on the broken/first subset. (iii) *Linked-name copying*: copy the name directly linked to the queried entity (present already at 0-shot, amplified by demos); mechanistically natural for induction-type heads, testable by logit attribution. For the checkpoint sweep, the pre-registered discriminating measure is

$$\Delta(\text{ckpt}) = \text{Acc}_{0\text{-shot}}^{\text{broken/first}} - \text{Acc}_{12\text{-shot}}^{\text{broken/first}},$$

compared against the demo-benefit curve (schema mechanisms predict co-emergence with ICL use) and the two-hop curve (the “immature ability” account predicts shrinkage as two-hop stabilizes).

9 Known Limitations and Audit Status

- `paired_base_id` references omit the order suffix and do not match result ids; the shipped paired-metric files are therefore empty. Order-pair analysis was computed independently via

`option_pair_id` (used for all both-orders numbers above); the field should be repaired in the next data revision.

- In 4-/12-shot runs the two order variants receive different sampled demonstrations; the both-orders metric tolerates this, but single-order contrasts do not isolate fact order alone.
- v3.1 lacks a direct+reorder control (is *one-hop* layout-robust?); planned for the next revision, alongside a containment-format fix.
- Broken-cell sample sizes ($n=34$ pairs) limit precision; the distractor-placement generator uses Python’s `hash()`, so the exact generated files (preserved, hashed in manifests) are the reproducibility anchor.
- Cross-model comparison relies on the dual-tokenizer-matched dataset; legacy Pythia numbers come from the original v3.1 files (40 of 100 families textually identical; the 60 renamed families preserve structure, ids, and balance by construction).

10 Lessons and Next Steps

Lessons: validate the task before interpreting the model (three generations of confounds were caught by built-in baselines, order pairs, and an independent logic verifier); error *anatomy* distinguishes mechanisms where accuracy alone cannot (identical broken/first accuracy, opposite failure modes); pre-registered rules work, including honestly reporting a partial pass and a justified deviation; tokenizer-dependent scoring must be enforced in code, not by convention.

Next: (1) costing dry-run on the OLMo final checkpoint (doubles as the patching tooling shakedown); (2) component screening — RI with nulls + logit attribution on the three-way contrast; (3) activation/path patching on the three pre-registered contrasts: clean vs broken/first, canonical vs reorder, 0-shot vs 12-shot broken/first; (4) the demo-composition behavioral experiment; (5) circuit extraction + faithfulness on held-out families; (6) checkpoint sweep with Δ , demo-benefit, and two-hop curves; (7) run the Pythia controls file; repair `paired_base_id`.

A Source Files

Paths relative to `final proj/project`; raw prompts and results are preserved alongside summaries. Repo: https://github.com/Maxgolu/NLP_proj.

- Proposal: `../NLPProjects_Proposal.pdf`; papers: `../papers/`; plan: `data_pipeline_plan.tex`.
- Runner (multi-model) and locks: `pilot_v2/pilot.py`, `parallel.py`, `model_lock.json`, `model_lock_olmo2.json`; OLMo operations and decision rule: `pilot_v2/README_OLMO2.md`; preflight: `pilot_v2/preflight_olmo2.py`.
- Data: generator `pilot_v3/build_data_v3.py`; OLMo adaptation + audits: `pilot_v3/adapt_data_olmo2.py`, `pilot_v3/audit_tokenizer.py`, `pilot_v3/verify_logic.py`, `pilot_v3/data_olmo2/` (incl. `adapt_audit_olmo2.json`).
- Yes/No era: `results/full_4shot_review/`, `results/full_sdpa_review/`, `results/prompt_v3_review/`; completion-v1: `results/completion_review/`, audit results: `results/completion_analysis/`; v2 (negative result): `results/completion_v2_*`.
- v3/v3.1 (Pythia): `results/completion_v3_4shot_v1/`, `results/completion_v3_12shot_v1/`, `results/completion_v3_1_0shot/`, `results/completion_v3_1_4shot/`, `results/completion_v3_1_12shot/`.

- OLMo comparison: `results/olmo2_results.tar.gz` (runs `olmo2_7b_v3_1_0shot`, `olmo2_7b_v3_1_4shot`, `olmo2_7b_v3_1_12shot`, `olmo2_7b_v3_1_controls`); analysis and report: `results/olmo2_vs_pythia/`.

References

- [1] Jie Ren et al. (2024). *Identifying Semantic Induction Heads to Understand In-Context Learning*. Findings of ACL, 6916–6932. <https://aclanthology.org/2024.findings-acl.412/>
- [2] Catherine Olsson et al. (2022). *In-context Learning and Induction Heads*. Transformer Circuits Thread. <https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html>
- [3] Nelson Elhage et al. (2021). *A Mathematical Framework for Transformer Circuits*. Transformer Circuits Thread. <https://transformer-circuits.pub/2021/framework/index.html>
- [4] Kevin Wang et al. (2022). *Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 Small*. <https://arxiv.org/abs/2211.00593>
- [5] Stefan Heimersheim and Neel Nanda (2024). *How to Use and Interpret Activation Patching*. <https://arxiv.org/abs/2404.15255>
- [6] Team OLMo et al. (2024). *2 OLMo 2 Furious*. <https://arxiv.org/abs/2501.00656>